JURONG JUNIOR COLLEGE
2014 JC 2 PRELIMINARY EXAMINATION
Higher 2

CHEMISTRY 9647/01

Paper 1 Multiple Choice

18 September 2014

1 hour

Additional Materials: Multiple Choice Answer Sheet
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name, class and exam index number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** or **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
The use of an approved scientific calculator is expected, where appropriate.

This document consists of **17** printed pages and **1** blank page.

Section A

For each question there are four possible answers, **A**, **B**, **C** and **D**. Choose the **one** you consider to be correct.

1 An acidified solution of hydrogen peroxide, H_2O_2, was oxidised by 0.05 mol dm^{-3} potassium manganate(VII) solution. The volume of oxygen gas produced at room temperature and pressure was 58 cm^3.

What is the volume of potassium manganate(VII) solution used in the oxidation of H_2O_2?

A 19.3 cm^3

B 20.7 cm^3

C 48.3 cm^3

D 120.8 cm^3

2 10 cm^3 of gaseous hydrocarbon was completely burnt in 80 cm^3 of oxygen (present in excess) at 425 K. After cooling to room temperature, the volume of the gaseous mixture decreased to 55 cm^3. A further reduction of 40 cm^3 was observed when the residual gas was passed through calcium hydroxide. All gas volumes were measured at room temperature and pressure.

What is the formula of the hydrocarbon?

A C_3H_8 **B** C_4H_8 **C** C_4H_{10} **D** C_5H_{12}

3 *Use of the Data Booklet is relevant to this question.*

The ion $\mathbf{X}^+$ has 54 electrons and 78 neutrons.

Which of the following statements is **incorrect**?

A Element **X** is isoelectronic with Xe.

B The first ionisation energy of element **X** is lower than that of Na.

C The oxide of **X** formed is expected to have a lower melting point than Na_2O.

D In an electric field, the ion $\mathbf{X}^+$ will be deflected at a smaller angle than that of Na^+.

4 Bromine trifluoride is a liquid at room temperature. It shows an electrical conductivity high enough to indicate that some auto-ionisation occurs, which is usually represented by the following equation:

$$2BrF_3 \;=\; BrF_2^+ \;+\; BrF_4^-$$

Which of the following is correct regarding BrF_3, BrF_2^+ and BrF_4^-?

A BrF_3 is planar while BrF_4^- is non-planar.

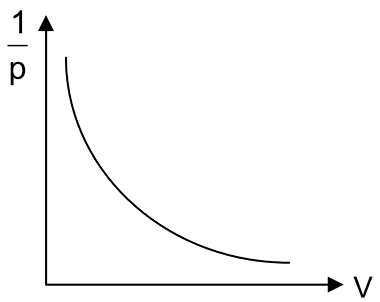
B BrF_2^+ is linear while BrF_4^- is tetrahedral in shape.

C The F–Br–F bond angle in BrF_3 is about the same as that in BrF_4^-.

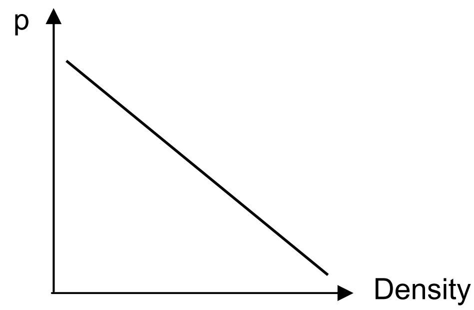
D There are more lone pairs of electrons around the Br atom in BrF_3 than that in BrF_2^+.

5 Which of the following diagrams correctly describes the behavior of a fixed mass of an ideal gas at constant *T*? (*T* is measured in K.)

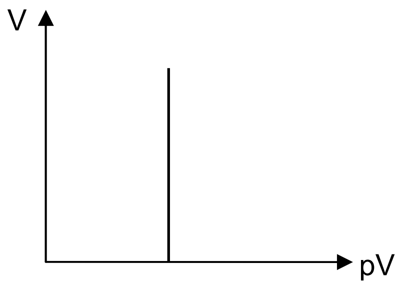
A



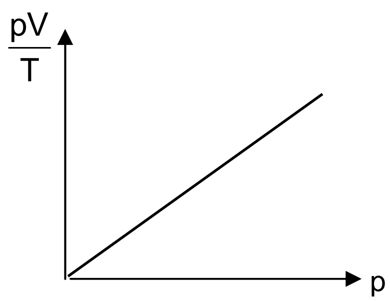
C



B



D



6 The enthalpy changes for two reactions are given by the equations:

$$Fe_2O_3(s) + 3CO(g) \rightarrow 2Fe(s) + 3CO_2(g) \qquad \Delta H = -26.8 \text{ kJ mol}^{-1}$$

$$FeO(s) + CO(g) \rightarrow Fe(s) + CO_2(g) \qquad \Delta H = -16.5 \text{ kJ mol}^{-1}$$

What is the enthalpy change for the reaction $Fe_2O_3(s) + CO(g) \rightarrow 2FeO(s) + CO_2(g)$?

A –59.8 kJ mol^{-1}

B –43.3 kJ mol^{-1}

C +6.2 kJ mol^{-1}

D +10.3 kJ mol^{-1}

7 Amyloids are insoluble fibrous protein aggregates sharing specific structural traits and are linked to various neurodegenerative diseases. The folding of amyloids into specific shapes under physiological condition is a spontaneous process.

What are the correct signs of ΔG, ΔH and ΔS for the folding process?

	ΔG	ΔH	ΔS
A	+	+	+
B	−	−	−
C	−	+	−
D	−	−	+

8 When PCl_5 was left to dissociate according to the equation,

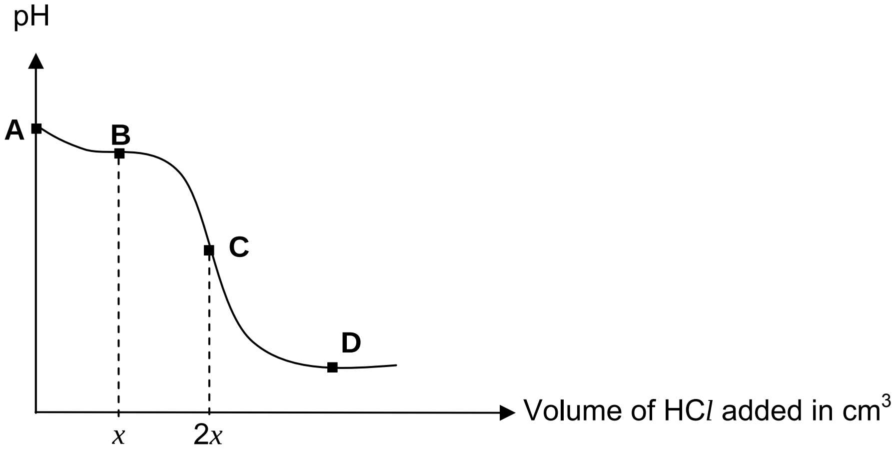
$$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$$

the partial pressure of PCl_5 at equilibrium was found to be 0.25 atm.

Given that the equilibrium constant, K_p, is 6.25 atm, what was the initial pressure of PCl_5 before dissociation?

A 1.25 atm **B** 1.50 atm **C** 2.00 atm **D** 2.50 atm

9 When 25.0 cm^3 of a 0.100 mol dm^{-3} solution of NH_3 was titrated with a 0.200 mol dm^{-3} HCl solution, the following titration curve was obtained.



Which of the following statements is correct about this titration?

A The K_b for the NH_3 solution is 1.58×10^{-5} mol dm^{-3} if the pH at point **A** is 11.1.

B The volume of HCl at point **B** is 12.5 cm^3.

C The pH at equivalence point **C** is above 7.

D A buffer mixture is formed at point **D**.

10 A researcher accidentally mixed 50.0 cm^3 of 0.100 mol dm^{-3} $MgSO_4$ solution with 50.0 cm^3 of 0.100 mol dm^{-3} $FeSO_4$ solution in the laboratory. It is suggested that the metal ions can be separated through selective precipitation by adding solid sodium hydroxide.

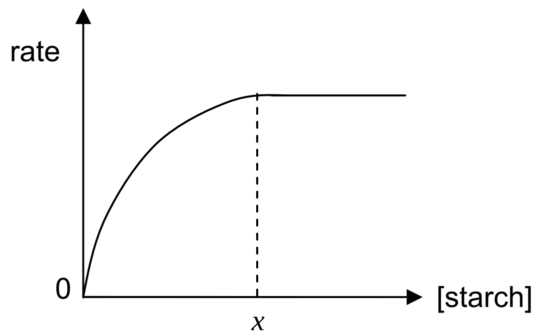
At 298 K, the solubility product of $Mg(OH)_2$ is 1.8×10^{-11} mol^3 dm^{-9} and that of $Fe(OH)_2$ is 4.1×10^{-15} mol^3 dm^{-9}.

What would be the pH of the resultant solution when maximum separation has occurred?

A 7.31 **B** 7.46 **C** 9.13 **D** 9.28

11 Amylase is the first enzyme discovered and isolated. It acts as a catalyst in the hydrolysis of starch.

In a single experiment, the rate of hydrolysis of starch was monitored as the reaction proceeded and the following graph was obtained.



Which statement about the reaction is **not** correct?

A When [starch] is smaller than x, the rate changes as [starch] changes.

B When [starch] is larger than x, the active sites of amylase are fully occupied.

C The order of reaction with respect to starch is constant at all concentrations.

D Throughout the experiment, [amylase] remains constant as it is not used up.

12 *Use of the Data Booklet is relevant to this question.*

Iodine and iodide ions undergo the following equilibrium in aqueous solution:

$$I_2(aq) + I^-(aq) \rightleftharpoons I_3^-(aq) \text{ ------- equilibrium 1}$$

I_2 is much more soluble in hexane than in water.

A cell comprises of a standard I_3^-/ I^- half–cell and a standard Cu^{2+}/Cu half–cell.

The electrode reaction for the standard I_3^-/I^- half–cell is

$$I_3^-(aq) + 2e^- \rightleftharpoons 3I^-(aq) \qquad E^\ominus = +0.536 \text{ V}$$

Which of the following statements is **not** correct?

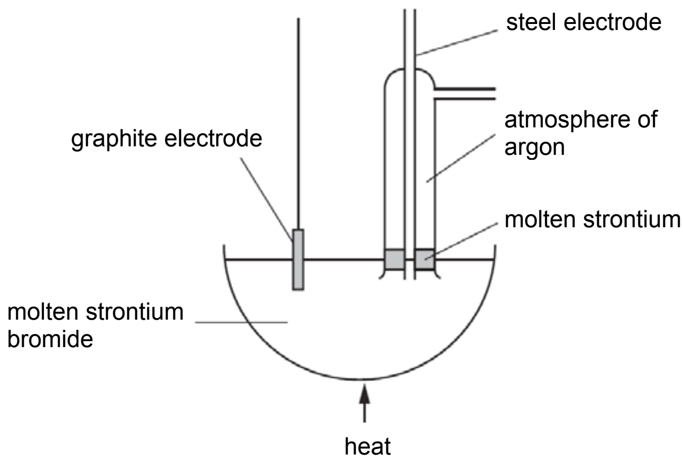
A The standard cell potential is +0.196 V.

B The addition of NaOH(aq) to the Cu^{2+}/Cu half–cell causes the cell potential to be more positive.

C The addition of hexane to the I_3^-/I^- half–cell causes the position of equilibrium 1 to shift to the left.

D The addition of hexane to the I_3^-/I^- half–cell causes the cell potential to be more positive.

13 Strontium metal can be obtained by the electrolysis of molten strontium bromide, $SrBr_2$, using the apparatus shown in the diagram.



Which statement concerning the electrolysis is correct?

A 193 C of electricity are needed to deposit 1 × 10^{-3} mol of strontium atoms at the anode.

B The region around the graphite electrode turns from red-brown to colourless.

C Without the atmosphere of argon, strontium hydroxide would be formed.

D An equal amount of moles of bromine and strontium are formed.

14 Compounds of Period 3 elements dissolve in water to form aqueous solutions that are acidic, basic or neutral.

Which of the sequence shows the order of **increasing** pH of the resultant solutions formed when the following compounds are dissolved in water?

A $NaCl$, $MgCl_2$, $SiCl_4$

B Al_2O_3, MgO, SO_2

C $AlCl_3$, $SiCl_4$, PCl_5

D P_4O_{10}, SiO_2, MgO

15 $BaCO_3$ decomposes at 1400 °C to produce CO_2 and BaO.

$BaSO_4$ decomposes at 1600 °C to produce SO_3 and BaO.

Which statement **best** explains the greater thermal stability of $BaSO_4$?

A The CO_2 molecule is smaller than SO_3.

B The CO_3^{2-} ions are more easily polarised than SO_4^{2-}.

C The charge density of CO_3^{2-} is greater than that of SO_4^{2-}.

D The lattice energy of $BaCO_3$ is more exothermic than $BaSO_4$.

16 The solubility of the silver halides in aqueous ammonia decreases from AgC*l* to AgI.

What helps to explain this trend?

A As a more powerful ligand, NH_3 can displace Cl^- ions and Br^- ions, but not I^- ions.

B Cl^- ions and Br^- ions form complexes with NH_3(aq), but I^- ions do not.

C The value of the solubility product of the silver halides decreases from AgC*l* to AgI.

D The covalent bonding between Ag and the halogen atom increases in strength from AgC*l* to AgI.

17 Sulfur is converted to SF_6 by fluorine, to SCl_2 by chlorine and to S_2Br_2 by bromine.

Which trend does this information best provide evidence for?

A the trend in bond energy: $Cl_2 > Br_2 > F_2$

B the trend in electronegativity: F > C*l* > Br

C the trend in first ionisation energy: F > C*l* > Br

D the trend in oxidising ability: $F_2 > Cl_2 > Br_2$

18 Which statement concerning transition metals is correct?

A They are the only metals of which the anhydrous chlorides have covalent bonds.

B They are the only metals which give coloured ions in an aqueous solution.

C They are the only metals which have more than one oxidation state.

D They are the only metals with a complete 4s orbital in their atoms.

19 Samples of the gases CH_3F and Cl_2 are mixed together and irradiated with light.

Which compound is produced in trace amounts by a termination stage in the chain reaction?

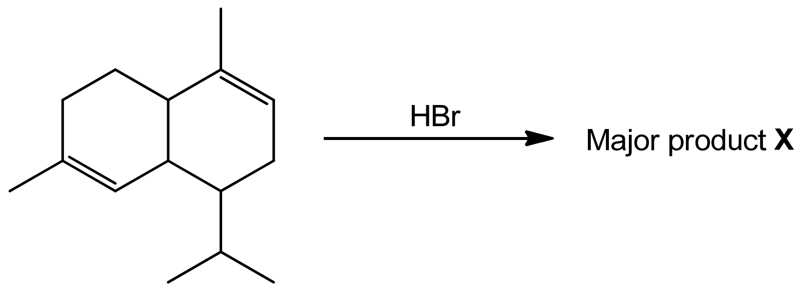
A HCl

B $CH_2{=}CH_2$

C CH_2FCH_2F

D CH_3CH_3

20 α-Cadinene is present in the oil derived from the plant juniper woods. **X** is the major compound when α-Cadinene is reacted with HBr in an inert organic solvent.



How many chiral carbons does **X** have?

A 3 **B** 4 **C** 5 **D** 6

21 Which reaction will **not** give propene as the major product?

A adding an excess of concentrated sulfuric acid to propan-1-ol, followed by heating

B adding warm aqueous sodium hydroxide to 2-bromopropane

C adding warm ethanolic sodium hydroxide to 1-bromopropane

D passing propan-2-ol vapour over heated aluminium oxide

22 Which property does benzene have as a consequence of the delocalisation of electrons in the benzene molecule?

A Benzene is a good conductor of electricity.

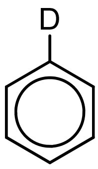
B Substitution in benzene takes place at one particular carbon atom.

C Addition reactions of benzene take place more easily than substitution.

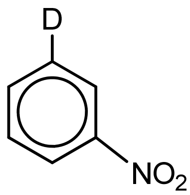
D The carbon-carbon bond lengths are between those of C–C bonds and C=C bonds.

23 Deuterium, D, is a heavy isotope of hydrogen. Deuteriobenzene is reacted with a mixture of nitric acid and sulfuric acid under controlled conditions, so that only mononitration takes place.

Assuming that the carbon-deuterium bond is broken as easily as a carbon-hydrogen bond, which proportion of the nitrated products will be 3-nitrodeuteriobenzene?



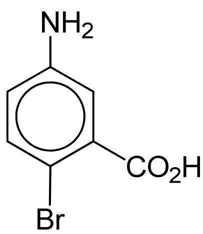
deuteriobenzene



3-nitrodeuteriobenzene

A 16% **B** 20% **C** 33% **D** 45%

24 A student wishes to synthesise compound **Y** from methylbenzene.



Y

Which of the following routes would give the maximum yield of **Y**?

A Oxidation → nitration → bromination → reduction

B Oxidation → nitration → reduction → bromination

C Bromination → nitration → reduction → oxidation

D Bromination → oxidation → nitration → reduction

25 Under the Montreal Protocol, the manufacture of chlorofluorocarbons has been phased out and they are being replaced by fluorocarbons.

One chlorofluorocarbon which was widely used as a solvent is CCl_2FCClF_2. Large stocks of CCl_2FCClF_2 is used up by converting it into the fluorocarbon CH_2FCF_3 by the following route.

$$CCl_2FCClF_2 \xrightarrow{\text{step 1}} CCl_3CF_3 \xrightarrow{\text{step 2}} CCl_2FCF_3 \xrightarrow{\text{step 3}} CH_2FCF_3$$

Which type of reaction is step 2?

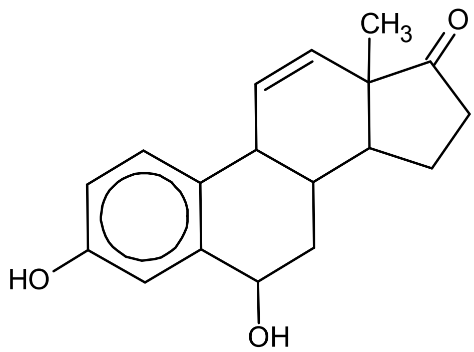
A isomerisation

B free radical reduction

C electrophilic substitution

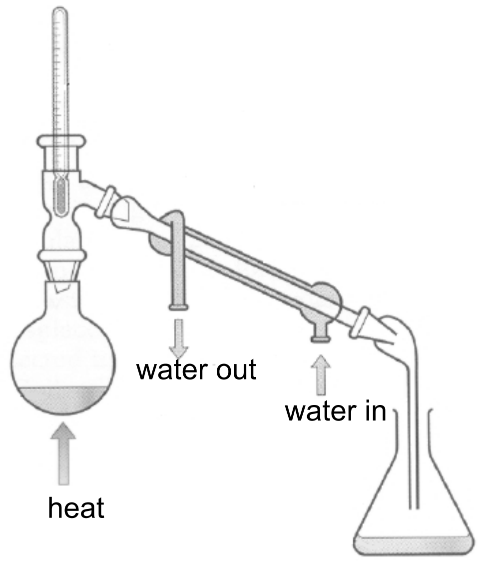
D nucleophilic substitution

26 After heating acidified potassium dichromate(VI) with the molecule below, how many sp^2 carbons are present in the resultant compound?



A 4 **B** 6 **C** 8 **D** 10

27 Which of the following compounds can be prepared by the apparatus shown?



A 1,2-dibromoethane, by using bromine and ethene

B Ethanoic acid, by using ethanol, sodium dichromate(VI) and sulfuric acid

C Bromoethane, by using ethanol, sodium bromide and concentrated sulfuric acid

D Phenyl benzoate, by using phenol and benzoyl chloride

28 An organic compound has the following properties:

- It gives a yellow precipitate with alkaline aqueous iodine solution.
- It gives an orange precipitate with 2,4-dinitrophenylhydrazine.
- It forms a brick-red precipitate with Fehling's solution.

Which compound could give these results?

A $CH_3CH(OH)CH_2CH_2CHO$

B $CH_3COOCH_2CH_2CHO$

C $CH_3COCH_2COCH_2CH_3$

D $CH_3COCH_2C_6H_4CHO$

29 In which pair of reactions would the intermediate be the same?

Reaction **I**: $CH_3CH_2CH_3 \rightarrow$ intermediate $\rightarrow (CH_3)_2CHCN$

Reaction **II**: $CH_3CH(OH)CH_3 \rightarrow$ intermediate $\rightarrow (CH_3)_2C(OH)CN$

Reaction **III**: $CH_3CH{=}CH_2 \rightarrow$ intermediate $\rightarrow (CH_3)_2CHNH_2$

Reaction **IV**: $CH_3CO_2CH_2CH_2CH_3 \rightarrow$ intermediate $\rightarrow CH_3CH_2CH_2Br$

A **I** and **II**

B **I** and **III**

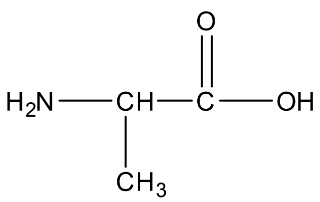
C **II** and **IV**

D **III** and **IV**

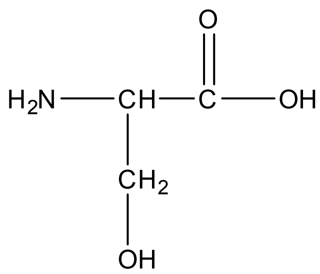
30 The secondary structure of a protein is held together by hydrogen bonds.

Which of the following amino acids would **not** allow the formation of hydrogen bonds in the structure of an α-helix?

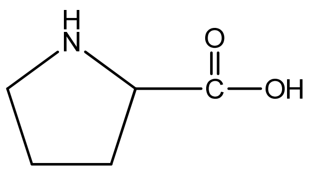
A



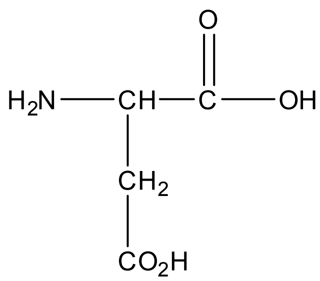
C



B



D



Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, **2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

No other combination of statements is used as a correct response.

31 *Use of the Data Booklet is relevant to this question.*

Species containing one or more unpaired electrons are said to be paramagnetic as they can be attracted by an external magnetic field.

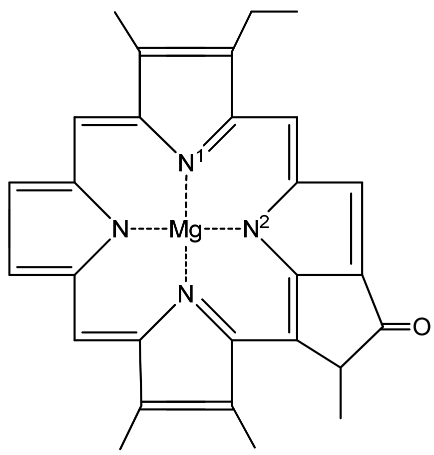
Which of the following species are **not** paramagnetic?

1 Cu^{+}

2 Fe^{2+}

3 Cr^{3+}

32 A simplified structure of a molecule of chlorophyll is shown. Mg ion is situated in the centre of a planar arrangement of the four N atoms.



Which of the following statements are correct?

1 Co-ordinate bond forms between Mg and N^1.

2 Ionic bond forms between Mg and N^2.

3 The hybridisation about N^1 is sp^2.

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, **2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

No other combination of statements is used as a correct response.

33 The rate of reaction between bromine and methanoic acid is first order with respect to bromine and to methanoic acid.

$$Br_2(aq) + HCO_2H(aq) \rightarrow 2Br^-(aq) + 2H^+(aq) + CO_2(g)$$

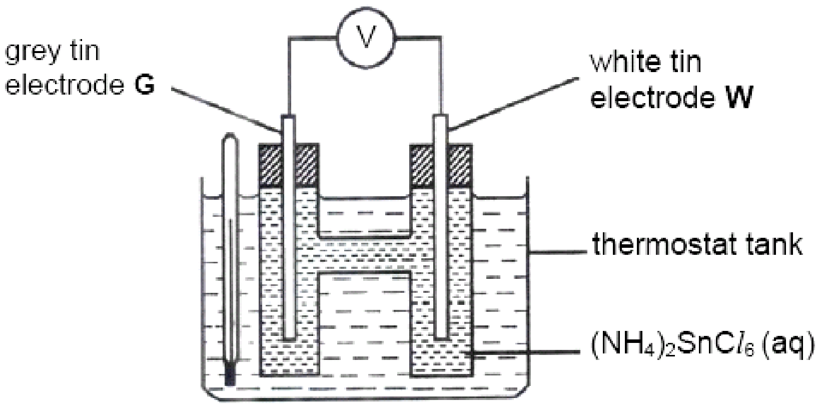
Which of the following can be correctly deduced from this information?

1 Doubling the concentration of methanoic acid doubles the rate of evolution of gas.

2 The half–life of the reaction is constant when $[Br_2]$ doubles.

3 Halving the concentration of both reactants simultaneously will halve the reaction rate.

34 The diagram shows an apparatus to find the transition temperature (18 ^oC) at which white tin and grey tin are in equilibrium. Below 18 ^oC, white tin dissolves from **W** and is deposited on **G** as grey tin.



Which of the following statements are correct?

1 The stable form of tin at 25 ^oC is grey.

2 Below 18 ^oC, electrons flow through the external circuit from **W** to **G**.

3 At 18 ^oC, no current flows.

35 Radium is the last member of the Group II elements.

Which statements are correct for radium?

1 Radium reacts vigorously with water, liberating hydrogen.

2 Radium is the strongest reducing agent in Group II.

3 Radium hydroxide is insoluble in water.

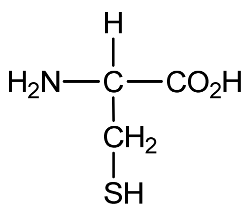
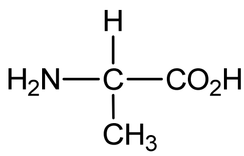
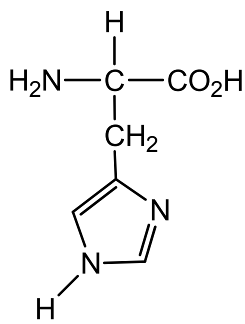
36 Which of the following properties are the same in both the optical isomers of lactic acid, $CH_3CH(OH)COOH$?

1 density

2 boiling point

3 standard enthalpy change of formation

37 A tripeptide, cys-ala-his, was analysed using electrophoresis. The tripeptide was hydrolysed and the resulting solution was added to a buffer solution of pH 6 which was then placed at the centre of the plate. A potential difference was then applied across the plate.

Amino acid	 cysteine (cys)	 alanine (ala)	 histidine (his)
Isoelectric point	5.05	6.00	7.60

With reference to the table above, which of the following statements are correct?

1 Alanine would remain at the centre of the plate.

2 Cysteine would migrate towards the anode.

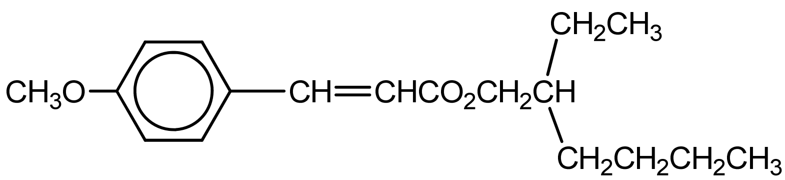
3 Histidine would migrate towards the cathode.

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, **2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

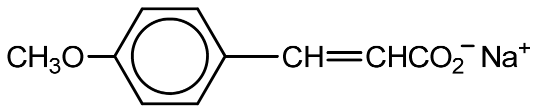
No other combination of statements is used as a correct response.

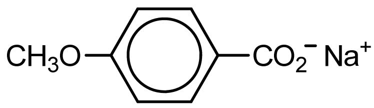
38 A sun protection cream contains the following ester as its active ingredient.



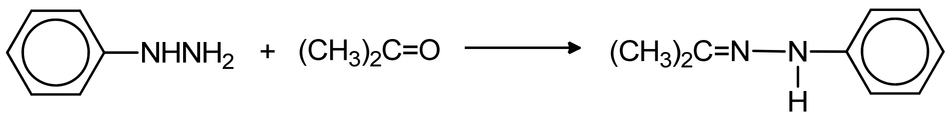
What are the products of its partial or total hydrolysis by aqueous sodium hydroxide?

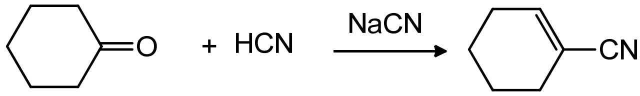
1 $CH_3CH_2CH_2CH_2CH(CH_2CH_3)CH_2OH$

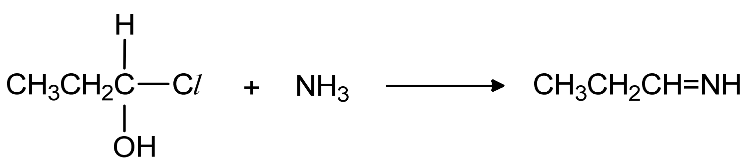
2 

3 

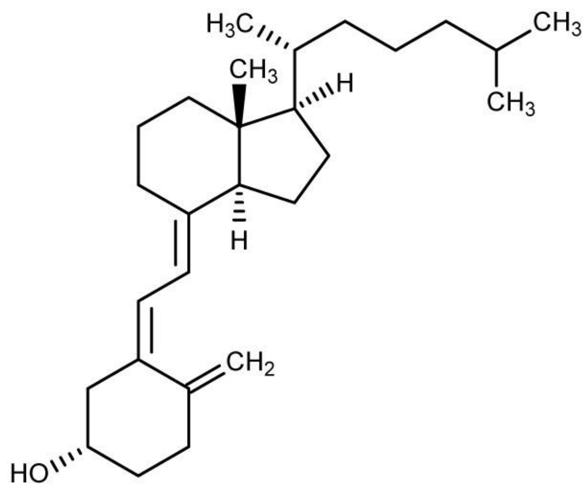
39 Which transformations involve a nucleophilic attack, followed by a dehydration?

1 

2 

3 

40 Cholecalciferol, also known as Vitamin D3, is generated in the skin of animals when light energy is absorbed.



Which statements are correct?

1 Cholecalciferol exhibits geometric isomerism.

2 Cholecalciferol reacts with ethanoyl chloride to yield a sweet-smelling product.

3 1 mole of cholecalciferol produces 3 moles of carbon dioxide when treated with hot acidified potassium manganate(VII).

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